TODO: Paper Title

Anonymous author
Anonymous affiliation

Abstract

Background. TODO: 1–2 sentences on the problem and why it matters.
Aims. TODO: the research questions or specific objectives.
Method. TODO: dataset, models, approaches compared.
Results. TODO: headline numbers, ordered by RQ.
Conclusions. TODO: what this means and the takeaway for practitioners.

2012 ACM Subject Classification Software and its engineering → Software notations and tools; Computing methodologies → Natural language processing

Keywords and phrases issue classification, retrieval-augmented generation, fine-tuning, large language models, mining software repositories

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1 Introduction

2 Related Work

Early IRC studies employed data mining techniques and ML classifiers [1, 14, 9]. For example, Antoniol et al. [1] leveraged a naive Bayes classifier to distinguish bugs from other issue reports. However, these methods showed low performance due to their limited semantic understanding of the issue report text [10].

To overcome this, recent methods have adopted transformer models due to their strong contextual understanding [6, 4, 17, 12, 13]. These studies typically fine-tune the models on historical issues in a supervised setting. For instance, Izadi et al. [12] fine-tuned a RoBERTa [15] model to classify issues into *Bugs*, *Features*, or *Questions*. Similarly, Trautsch et al. [17] trained a seBERT [18] model to classify issues into the same three categories. Although transformer-based approaches achieve strong results, they rely on large amounts of project-specific data to be effective, hindering their generalizability on projects with a small number of labeled issues [10, 12]. Additionally, they struggle to accurately classify minority classes with fewer training examples [13, 6].

These challenges have motivated the exploration of LLMs, which exhibit strong text classification capabilities thanks to extensive pretraining on massive corpora. [10, 3, 2, 5, 19, 7]. Studies show that LLMs perform well in zero-shot settings [7, 2]. SOTA LLM-based methods achieved the best performance by fine-tuning the models on labeled datasets [10, 3]. For example, Aracena et al. [3] instruction-tuned several models on the NLBSE '23 and '24 datasets.

3 Approach

3.1 Problem Formulation

Given a query issue report, we classify it into only one of the three lables: *bug*, *feature*, or *question*. Only the issue's title and description texts are used for classification with no other signals from the issue tracker. The defined label space is adopted directly from prior IRC studies. [2, 3, 12, 14, 1].

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3.2 Indexing and Retrieval

To prepare the issues for retrieval based on textual similarity, we first embed their textual content (i.e., *title* + *description*). We do not embed the issue labels. Instead, we retain them as metadata associated with each embedded issue. To generate the embeddings, we use the `all-MiniLM-L6-v2` model from the Sentence-Transformers library [16]. We selected this model due to its lightweight architecture, fast performance, and effectiveness on issue report text [11]. After embedding all issue reports, we index them using the Faiss vector database library [8], which enables efficient high-dimensional similarity retrieval. At the retrieval stage, the query issue is first embedded with the same `all-MiniLM-L6-v2` model. Next, Faiss calculates the cosine similarity of indexed issue reports to the query, and returns the top-k most similar candidates.

4 Experimental Setup

5 RQ1 – Comparison

6 RQ2 – Diagnosis

7 RQ3 – Bridging

8 Discussion

9 Threats to Validity

10 Conclusion

11 Data Availability

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